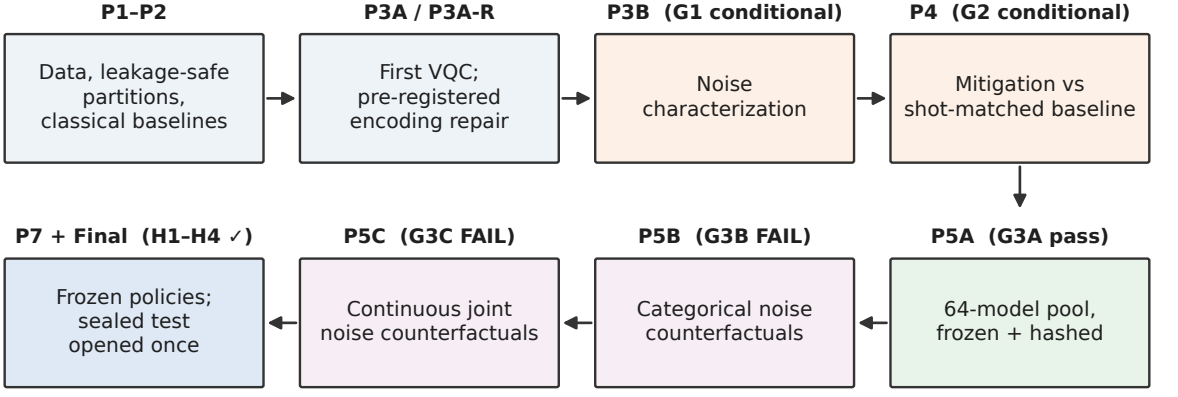

Every phase: pre-register (config + code) → push → clean-tree reported run → validate → commit and push.

Gates decide whether to continue; controller-test outcomes stay sealed until the final protocol is pushed.